

國立陽明交通大學

114 學年度碩士班考試入學招生試題

科目：線性代數與離散數學(8102)

系所班別：資訊聯招

考試日期：114 年 2 月 6 日 第 2 節

【不可使用計算機】

*作答前請先核對試題、答案卷(試卷)與准考證之
所組別與考科是否相符！！

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第 1 頁, 共 3 頁

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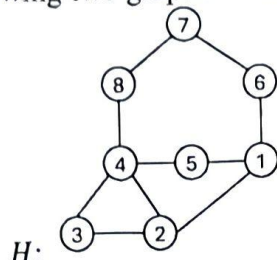
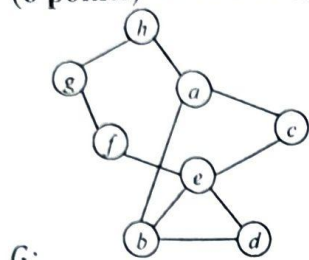
1. (11 points) Consider the classic Connect Four game, where players choose a color and then take turns dropping colored tokens into a six-row, seven-column vertically suspended grid. The objective of the game is to be the first to form a horizontal, vertical, or diagonal line of four of one's own tokens. Suppose that two players, one with yellow and the other with red tokens, are playing the game. Propositional functions $Y(i, j)$, $R(i, j)$, and $B(i, j)$ assert that the square in the i -th row and the j -th column (refer to the figure) contains a yellow, a red, and no token, respectively. **Please use these predicates together with the connectives permitted to translate the statements in (a) to (c) into logic expressions.** Permitted connectives are $\vee$, $\wedge$, $\neg$, $\rightarrow$, and the pair of parentheses " $()$ ". For any propositional function $f(x, y)$, you should use the notation $\bigvee_{i=m}^n f(i, j)$ for $f(m, j) \vee f(m+1, j) \vee \dots \vee f(n, j)$ and the notation $\bigwedge_{i=m}^n f(i, j)$ for $f(m, j) \wedge f(m+1, j) \wedge \dots \wedge f(n, j)$.

		columns						
		1	2	3	4	5	6	7
rows	1							
	2							
	3							
	4							
	5							
	6							

- (a) (4 points) Four squares in the 2nd row contains four red tokens in a row.
- (b) (4 points) Each column contains at least a yellow token.
- (c) (3 points) If the square in the i -th row and the j -th column contains a token, the square contains either a red token or a yellow token but not both.
2. (14 points)
- (a) (4 points) Given a domain D and two predicates P and Q , let R be the truth set of $P(x)$ and S be the truth set of $Q(x)$. Pick up all the following statements that are always true? ① $|R| \leq |D|$. ② If $\forall x \in D, P(x) \rightarrow Q(x)$, then $R \subseteq S$. ③ If $R \cap S = \emptyset$, then $P(x) \vee Q(x) = T$. ④ If $|R| \leq |S|$, then $\forall x \in D, P(x) \rightarrow Q(x)$. ⑤ If $R \subseteq S$, then $\forall x \in D, \neg P(x) \vee Q(x)$.
- (b) (10 points) Let $\aleph_0$ denote the cardinality of a countable infinite set. Name two infinite sets and use the inclusion-exclusion principle to show that $\aleph_0 + \aleph_0 = \aleph_0$ (Disregard whether the principle holds for infinite sets)
3. (12 points) Assigning 8 distinct tasks to 3 identical servers, where every task must be assigned to exactly one server.
- (a) (4 points) Suppose each server can handle any number of tasks, including none. How many ways can the 8 tasks be assigned to the 3 servers with no restrictions?
- (b) (2 points) Suppose that each server must handle at least one task. How many ways can the tasks be assigned?
- (c) (2 points) Suppose that each server must handle at least one task, and no server can handle 5 tasks or more, how many valid task assignments are possible?
- (d) (4 points) Now, consider that the servers are labeled (i.e., no longer identical), and each server must handle at least one task. Additionally, no server can handle 5 tasks or more. How many valid task assignments are there in this case?
4. (7 points) Consider the set $A = \{a, b, c, d, e\}$ and the relation R on A defined as follows: $R = \{(a, b), (b, c), (c, d), (d, e), (a, e)\}$.
- (a) (1 point) Is R reflexive?
- (b) (1 point) Is R symmetric?
- (c) (1 point) Is R transitive?
- (d) (4 points) Does R define a partial ordering? If not, what is the minimal number of ordered pairs that must be added to R to make it a partial order?

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5. (6 points) Consider the following two graphs
- G
- and
- H
- :



- (a) (2 points) Are graphs
- G
- and
- H
- isomorphic?

- (b) (4 points) If they are isomorphic, please provide the isomorphism function (a mapping from the vertices of
- G
- to the vertices of
- H
- that preserves adjacency). If they are not, explain in detail why they are not isomorphic.

6. (25 points) Answer the following problems. Any negative total score will be adjusted to 0 points.

Consider the problems (a)-(d), determine whether each statement is True or False. If a statement is False, provide a justification explaining why it is incorrect. You will receive 4 points for each correct answer, a penalty of -4 points for each incorrect answer, or 0 points for no answer.

For the problems (e)-(g), find the solutions and provide your answers. You will receive 3 points for each correct answer, or 0 points for each incorrect answer or no answer.

- (a) A square matrix A is invertible if and only if $\lambda = 1$ is an eigenvalue of A .
- (b) If W is a subspace of $\mathbb{R}^n$, then the orthogonal complement of W is also a subspace of $\mathbb{R}^n$.
- (c) If E is an $m \times m$ elementary matrix and A is an $m \times n$ matrix, then the null space of EA is the same as the null space of A .
- (d) If $f = \cos^2 x$, $g = \sin^2 x$, and $h = 1 + x^2$, then h lies in the space spanned by f and g .
- (e) Find the number of parameters in the general solution of $Ax = 0$ if A is a 5×7 matrix of rank 3.
- (f) Find the rank of a 5×6 matrix A for which $Ax = 0$ has a two-dimensional solution space.
- (g) Find the value of k such that the vector $(k, 21, -12)$ is a linear combination of vectors $(-2, 2, 1)$ and $(2, 1, -2)$.

7. (5 points) If
- $*$
- 's denote non-zero numbers, how many non-zero product terms in the expansion of the determinant

$$\begin{vmatrix} * & 0 & * & * & 0 & * \\ 0 & 0 & * & * & 0 & * \\ 0 & * & * & * & * & 0 \\ 0 & * & * & 0 & * & 0 \\ 0 & * & * & * & * & 0 \\ * & 0 & * & * & 0 & * \end{vmatrix}.$$

8. (10 points) Find the orthogonal matrix
- Q
- in which
- $q_{11} > 0$
- and
- $q_{22} > 0$
- , and the diagonal matrix
- D
- such that

$$\begin{bmatrix} x & y \end{bmatrix} Q D Q^T \begin{bmatrix} x \\ y \end{bmatrix} = 6x^2 + 24xy - y^2.$$

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9. (10 points) Find the matrix representation of the linear transformation

$$L\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 5x - 3y \\ x + y \\ -3x + 2y \end{bmatrix}$$

with respect to bases $B_1 = \left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$ and $B_2 = \left\{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}\right\}$.